

46 YEARS OF EXCELLENCE | SRAMAYEVA JAYATHE | NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR STAR - A & APEX, ALL IITS **DT: 11/01/2026**
Time: 3 Hrs **GTM – 15 (TBT-2)** **Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

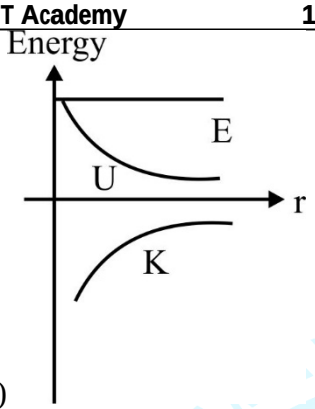
1. In the expression $P = EL^2m^{-5}G^{-2}$, where E, m, L and G denote energy, mass, angular momentum and gravitational constant, respectively. The dimensions of P are

(A) $\left[MLT^0\right]$ (B) $\left[M^2LT^{-1}\right]$ (C) $\left[M^0L^0T^0\right]$ (D) $\left[M^0LT^{-2}\right]$

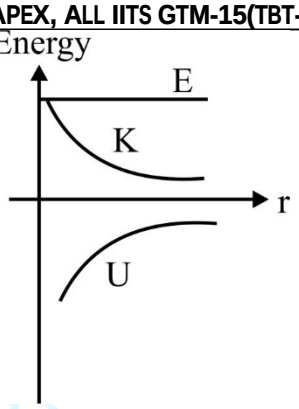
2. The correct graph representing the variation of total energy(E), kinetic energy(K) and potential energy (U) of a satellite with its distance from the centre of earth is

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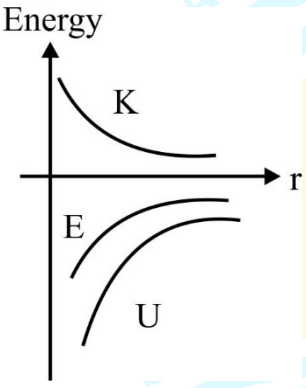
(A)



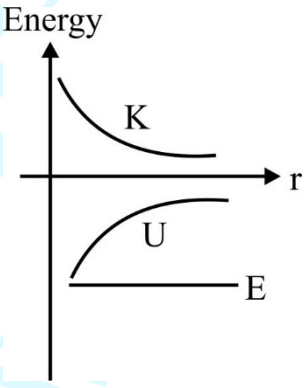
(B)



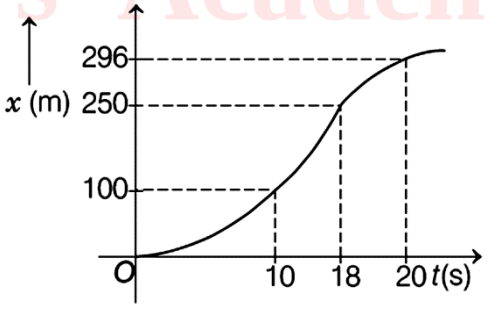
(C)



(D)



3. For motion of the car between $t = 18s$ and $t = 20s$ which of the given statement is correct?



(A) The car is moving in a positive direction with a positive acceleration.

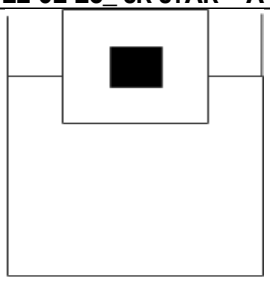
(B) The car is moving in a negative direction with a positive acceleration.

(C) The car is moving in positive direction with a negative acceleration.

(D) The car is moving in negative direction with a negative acceleration.

4. There is a metal cube inside a block of ice which is floating on the surface of water. The ice melts completely and metal falls in the water. Water level in the container.

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(A) rises (B) falls

(C) remains same (D) Nothing can be concluded

5. Consider the following statements about sound passing through a gas.

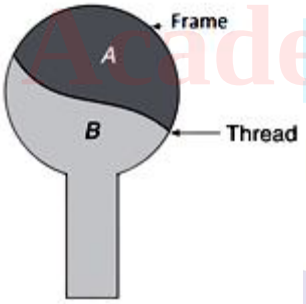
1) The pressure of the gas at a point oscillates in time.

2) The position of a small layer of the gas oscillates in time.

A) Both 1 and 2 are correct B) 1 is correct but 2 is wrong.

C) 2 is correct but 1 is wrong D) Both 1 and 2 are wrong

6. A thread is tied slightly looses to a wire frame as shown in the figure. And the frame is dipped into a soap solution and taken out. The frame is completely covered with the film. When the portion A is punctured with a pin, the thread:



(A) becomes convex towards A (B) becomes concave towards A

(C) remains in the initial position (D) take a straight line shape

7. A closed cubical box is completely filled with water and is accelerated horizontally towards right with an acceleration a. The resultant normal force by the water on the top of the box

A) passes through the cenre of the top

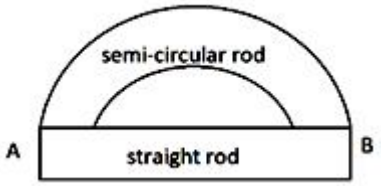
B) passes through a point to the right of the centre

C) passes through a point to the left of the centre

D) becomes zero

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8. Two rods (one semi-circular and other straight) of same material and of same cross-sectional area are joined as shown in figure. The points A and B are maintained at different temperatures. The ratio of the heat transferred through a cross-section of a semi-circular rod to the heat transferred through a cross-section of the straight rod in a given time is



(A) $2:\pi$ (B) $1:2$ (C) $\pi:2$ (D) $3:2$

9. An ideal gas is held in a container of volume V at pressure p. The average speed of a gas molecule under these conditions is v. If now the volume and pressure are changed to 2V and 2p, the average speed of a molecule will be

(A) v/2 (B) v (C) 2v (D) 4v

10. A ball is travelling with uniform translational motion. This means that

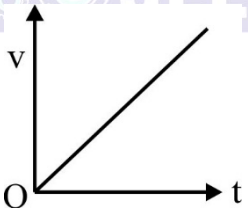
(A) it is at test

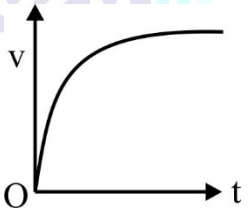
(B) the path can be a straight line or circular and the ball travels with uniform speed

(C) all parts of the ball have the same velocity (magnitude and direction) and the velocity is constant

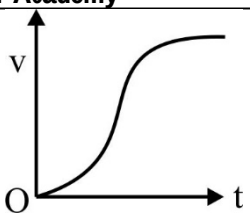
(D) the centre of the ball moves with constant velocity and the ball spins about its centre uniformly

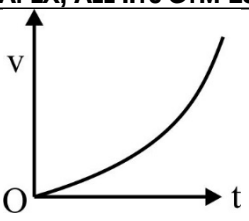
11. Which of the diagrams in figure correctly shows the velocity of an iron sphere falling freely in a lake having sufficient depth to impart it a terminal velocity?

(A)


(B)


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(C) 

(D) 

12. Math the Column I (rotaiton of different bodies) with Column II (their moment of inertia) and select the correct answer from the codes given below.

Column-I	**Column –II**
A) Circular disc of radius R about an axis passing through the diameter	1) $MR^2/4$
B) Hollow cylinder of radius R about its natural axis	2) MR^2
C) Solid cylinder of radius R about its natural axis	3) $MR^2/2$
D) Solid sphere of radius R about an axis passing through its diameter.	4) $(2/5)MR^2$

A) $A-4; B-3; C-2; D-1$ B) $A-1; B-2; C-3; D-4$

C) $A-1; B-3; C-2; D-4$ D) $A-2; B-1; C-3; D-4$

13. A transvers pulse travelling in a rope is reflected at the boundary with another rope. If the reflected pulse is not inverted, then wavelength of transmitted pulse is

(A) longer than original pulse

(B) shorter than original pulse

(C) equal to original pulse

(D) information are not sufficient to answer

14. A vector may change, if

(A) frame of reference is translated (B) frame of reference is rotated

(C) vector is translated parallel to itself (D) vector is rotated

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15. STATEMENT – I: In an elastic collision of two brilliard balls, the total kinetic energy is conserved during the short time of colisions of the balls (i.e., when they are in contact)

STATEMENT – II: If the potential energy of two brilliard balls depends only on the separation of distance between their centres, then the collision is elastic

A) Statement – I and Statement – II are correct and Statement – II is explaining Statement – I

B) Statement – I and Statement – II are correct but Statement – II is not explaining Statement – I

C) Statement – I is incorrect and Statement – II is correct

D) Statement – I and Statement – II both are incorrect

16. **Assertion (A) :** Moment of inertia of a circular disc of mass 'M' and radius 'R' about X, Y axes (passing through its plane) and Z-axis which is perpendicular to its plane were found to be I_x, I_y and I_z respectively. The respective radii of gyration about all the three axes will be the same.

Reason (R) : A rigid body making rotational motion has fixed mass and shape.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) A is correct but R is not correct

(B) Both A and R are correct and R is the correct explanation of A

(C) Both A and R are correct but R is NOT the correct explanation of A

(D) A is not correct but R is correct

17. An ideal gas in a closed container is slowly heated. As its temperature increases, which of the following statements are true?

A. The mean free path of the molecules decreases

B. The mean collision time between the molecules decreases

C. The mean free path remains unchanged

D. The mean collision time remains unchanged

(A) C and D (B) A and D (C) B and C (D) A and B

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18. A satellite of mass M is in a circular orbit of radius R about the centre of earth. A meteorite (very small size) of the same mass, falling towards the earth collides with the satellite completely inelastically. The speeds of the satellite and the meteorite are the same, just before the collision. The subsequent motion of the combined body will be

(A) Such that it escapes to infinity (B) In an elliptical orbit or path

(C) In the same circular orbit of radius R

(D) In a circular orbit of a different radius

19. **Assertion (A) :** Clothes containing oil or grease stains cannot be cleaned by water wash.

Reason (R) : Because the angle of contact between the oil/grease and water is obtuse. In the light of the above statements, choose the correct answer from the option given below.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A)

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(C) (A) is true but (R) is false

(D) (A) is true but (R) is true

20. A sonometer wire is divided in many segments using bridges. If fundamental frequencies of the segments are $n_1, n_2, n_3, ...$, then the fundamental frequency (n) of entire sonometer wire will be (if the divisions were not made)

(A) $n = n_1 + n_2 + n_3 + ...$ (B) $n = \sqrt{n_1 \times n_2 \times n_3 \times ....}$

(C) $\frac{1}{n} = \frac{1}{n_1} + \frac{1}{n_2} + \frac{1}{n_3} + ...$ (D) None of the above

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SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

Note: For Questions 21 to 25, follow these rules while marking answers in the OMR sheet:

If only option 1 is correct, bubble "1". If options 1 and 2 are correct, bubble "12". If options 1, 2, and 3 are correct, bubble "123". If options 1, 2, 3, and 4 are correct, bubble "1234". If options 1 and 3 are correct, bubble "13". If options 2 and 4 are correct, bubble "24". If options 3 and 4 are correct, bubble "34"., and so on Bubble all correct option numbers together as a single answer.

21. Identify the correct statements from the following
1) Work done by a man in lifting a bucket out of a well by means of a rope tied to the bucket is negative
2) Work done by gravitational force in lifting a bucket out of a well by a rope tied to the bucket is negative.
3) Work done by friction on a body sliding down an fixed rough inclined plane is positive.
4) Work done by an applied force on a body moving on a rough horizontal plane with uniform velocity in zero
5) Work done by the air resistance on an oscillating pendulum in negative.

22. Every planet revolves around the sun in an elliptical orbit, Identify the correct statements from the following
1) The force acting on a planet is inversely proportional to square of distance from sun.
2) Force acting on planet is inversely proportional to product of the masses of the planet and the sun
3) The centripetal force acting on the planet is directed away from the sun.
4) The square of time period of revolutions of planet around sun is directly proportional to cube of semi-major axis of elliptical orbit.

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23. In a linear simple harmonic motion (SHM) , Identify the correct statements from the following

1) Restoring force is directly proportional to the displacement.

2) The acceleration and displacement are opposite in direction.

3) The velocity is maximum at mean position.

4) The acceleration is minimum at extreme points

24. Identify the correct statements from the following regarding a given mass of ideal gass.

1) The average kinetic energy of a gas molecule decreases when the temperature is reduced.

2) The average kinetic energy of a gas molecule increases with increase in pressure at constant temperature.

3) The average kinetic energy of a gas molecule decreases with increase in volume isothermally.

4) Pressure of a gas increases with increases in temperature keeping volume constant.

25. Identify the correct statements from the following

1) A horse cannot pull a cart and run in empty space because there is no external force in empty space.

2) Passengers are thrown forward from their seats when a speeding bus suddenly stops, due to inertia of motion

3) It is easier to push a lawn roller than to pull it

4) A cricketer moves his hands backwards while taking a catch, to reduce the rate of change of momentum and hence to reduce the force necessary to stop the ball.

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SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. Which of the following statement is incorrect?

A) $\lambda = \frac{h}{p}$ is valid for photon

B) Paschen and Brackett series of H – spectrum shows overlap with each other

C) Lymann series consist of electromagnetic radiation in U.V region.

D) Bohr's theory could explain well about the formation of covalent bonds and molecules by atoms

27. Which of the following statements of quantum mechanics was in agreement with Bohr's model?

(i) The path of an electron in an atom can never be determined accurately.

(ii) The energy of electrons in atom is quantized i.e., can only have specific values.

(iii) An orbital cannot contain more than two electrons.

(A) Only (i) (B) (i) and (ii) (C) Only (ii) (D) (ii) and (iii)

28. Which of the following statement(s) is/are incorrect regarding photoelectric effect?

(i) The number of electrons ejected is proportional to the intensity of light.

(ii) Photo electrons have higher kinetic energy compared to that of incident photon.

(iii) The kinetic energy of ejected electrons depends upon the brightness of light.

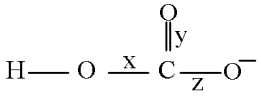
(iv) The kinetic energy of the ejected electron is proportional to the frequency of the incident radiation.

(A) (i) and (ii) (B) (ii) and (iii) (C) (iii) only (D) (ii) and (iv)

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29. Among the following, select the species that is isoelectronic and isostructural with CO_2

(A) N_3^- (B) $(CNO)^-$ (C) $(NCN)^{2-}$ (D) All of these

30. 

The relation between x, y and z in bicarbonate ion with respect to bond length is____

(A) x > y > z (B) x > z > y (C) (z = y) > x (D) x >(y= z)

31. Arrange in the increasing order of basic character : MgO, SrO, K_2O, NiO, Cs_2O

(A) MgO < SrO < NiO < K_2O < Cs_2O

(B) MgO < NiO < SrO < K_2O < Cs_2O

(C) MgO < SrO < K_2O < NiO < Cs_2O

(D) NiO < MgO < SrO < K_2O < Cs_2O

32. Bond order normally gives idea of stability of a molecular species. All the molecules viz. H_2, Li_2 and B_2 have the same bond order yet they are not equally stable. Their stability order is

(A) $H_2 > B_2 > Li_2$ (B) $Li_2 > H_2 > B_2$

(C) $Li_2 > B_2 > H_2$ (D) $B_2 > H_2 > Li_2$

33. Which of the following statements is/are not correct for Bonding combination of atomic orbitals?

(i) The combining atomic orbitals must have the same or nearly the same energy.

(ii) Greater the extent of overlap, the greater will be the electron density away from the nuclei of a moleculer orbital.

(iii) $2p_z$ orbital of one atom can combine with either of $2p_x$,$2p_y$ or $2p_z$ orbital of other atom as these orbitals have same energy.

(A) (i) and (ii) (B) (iii) only (C) (i) only (D) (ii) and (iii)

34. Which of the following are not state functions?

(I) q_{irr} (II) q_{rev} (III) w (IV) H – TS

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(A) (I) and (IV) (B) (II), (III) and (IV)

(C) (I), (II) and (III) (D) (II) and (III) only

35. Which of the following statements is/are correct ?

(i) The second ionization enthalpy of oxygen element is greater than that of fluorine element.

(ii) The third ionization enthalpy of phosphorus is greater than that of aluminium.

(iii) The first ionization enthalpy of aluminium is slightly greater than that of gallium.

(iv) The second ionization enthalpy of copper is greater than that of zinc.

(A) i,iv only (B) ii,ivonly (C) i,ii only (D) i,ii,iv

36. The standard enthalpy of formation $(\Delta_f H°_{298})$ for methane, CH_4 is -74.9 kJ mol^{-1}. In order to calculate the average energy given out in the formation of a C – H bond from this it is necessary to know which one of the following?

(A) The dissociation energy of only H_2 molecule

(B) The first four ionisation energies of carbon

(C) The dissociation energy of H_2 and enthalpy and sublimation of carbon (graphite)

(D) The first four ionisation energies of carbon and electron affinity of hydrogen

37. Which of the following statement(s) is/are correct for H_3PO_4

(i) 2nd and 3rd $\left(K_{a_2}, K_{a_3}\right)$ ionization constants are smaller than the first ionisation $\left(K_{a_1}\right)$

(ii) PO_4^{3-} is a conjugate base of H_3PO_4

(A) Both (i) and (ii) (B) Neither (i) nor (ii) (C) Only (i) (D) Only (ii)

38. When hydrochloric acid is added to cobalt nitrate solution at room temperature, the following reaction takes place out the reaction mixture becomes blue. On cooling the mixture it becomes pink. On the basis of this information mark the correct answer.

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$[Co(H_2O)_6]^{2+}(aq) + 4Cl^-(aq) \rightleftharpoons [CoCl_4]^{2-}(aq) + 6H_2O(\ell)$

(A) $\Delta H > 0$ for the reaction

(B) $\Delta H < 0$ for the reaction

(C) $\Delta H = 0$ for the reaction

(D) The sign of ΔH cannot be predicted on the basis of this information

39. In HS^{-1}, I^-, RNH_2 and NH_3, order of proton accepting tendency will be

(A) $I^- > NH_3 > RNH_2 > HS^-$ (B) $HS^- > RNH_2 > NH_3 > I^-$

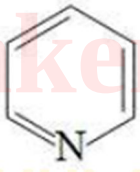
(C) $RNH_2 > NH_3 > HS^{-1} > I^-$ (D) $NH_3 > RNH_2 > HS^- > I^-$

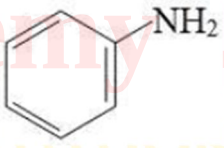
40. Which among the following purification methods is based on the principle of "Solubility" in two different solvents?

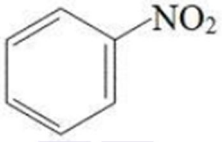
(A) Distillation (B) Differential Extraction

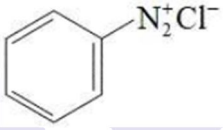
(C) Column Chromatography (D) Sublimation

41. Which of the following compounds will be suitable for Kjeldahl's method for nitrogen estimation?

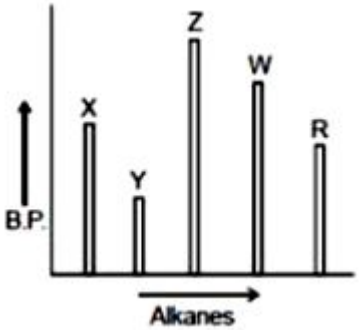
(A) 

(B) 

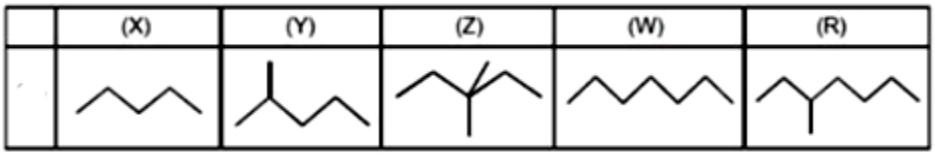
(C) 

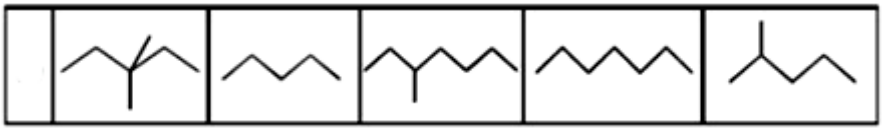
(D) 

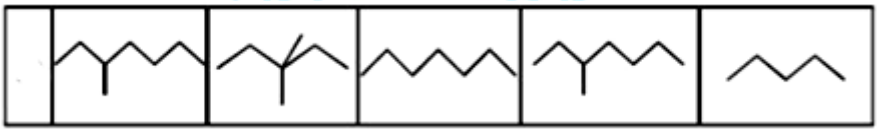
42. Observe the following diagram for relative boiling points of different alkanes. Identify the hydrocarbons X, Y, Z, W, R.

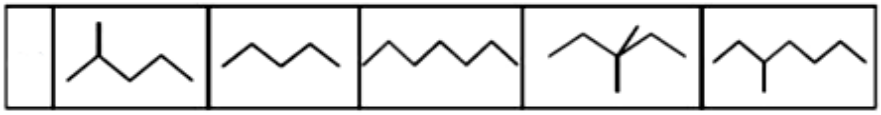


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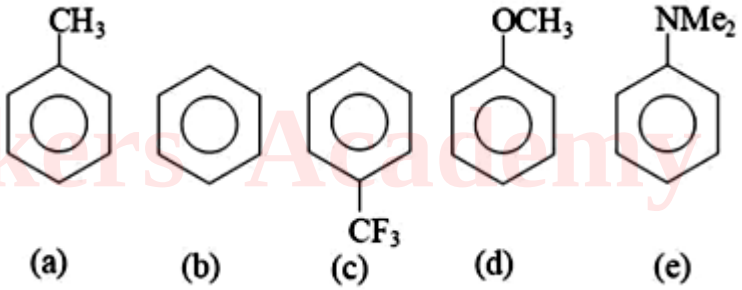
(A) 

(B) 

(C) 

(D) 

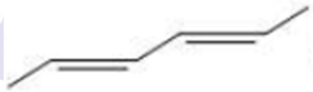
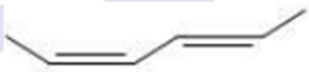
43. Decreasing order of reactivity towards electrophilic substitution for the following compounds is

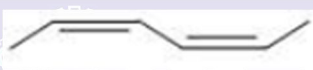


(A) a > d > e > b > c (B) d > a > e > c > b

(C) e > d > a > b > c (D) c > b > a > d > e

44. Which of the following is correct order for heat of combustion?

(A)  (B) 

(C) 

(A) C > B > A (B) A > B > C (C) B > A > C (D) C > A > B

45. Which of the following statement is/are incorrect ?

(i) During free expansion of ideal gas in vacuum , the entropy change of surrounding is negative

(ii) Free expansion of ideal gas results in decrease in internal energy

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(iii) The work done by the gas in irreversible isothermal expansion is more than the work done by the gas in a reversible isothermal expansion for the same change in volume

(A) I only (B) ii,iii only (C) i,ii only (D) i,ii,iii

SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

46. How many of the following statements are correct

1) The electrical discharge through the gases could be observed only at very high pressures and at very high voltage in cathode ray tube.

2) The characteristics of anode rays do not depend upon the material of electrodes and the nature of the gas present.

3) For a charged particle, greater the charge and lesser is the mass, greater is the deflection upon interaction with the electric or magnetic field.

4) Bohr model of the hydrogen atom, only ignores dual behaviour of matter but conforms with Heisenberg uncertainty principle.

5) Wave nature of the electromagnetic radiation could explain variation of heat capacity of solids as a function of temperature.

47. How many of the following statements are correct

1) In the titration of $KMnO_4$ with oxalic acid, $KMnO_4$ acts as self indicator.

2) In permanganate titration, HCl and HNO_3 are not used but H_2SO_4 is used for acidification.

3) In the titration of I_2 and $Na_2S_2O_3$ the oxidizing agent is I_2

4) Oxalic acid and Mohr's salt can be used as primary standards in volumetric analysis but not $KMnO_4$

5) Molality and mole fraction are temperature independent concentration terms for solid – liquid binary solutions

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48. How many of the following statements are correct

1) Dipolemoment: $BF_3 < NF_3 < NH_3$

2) PCl_3F_2 is non-polar and PF_3Cl_2 is polar

3) SO_3 is non – polar and SO_2 is polar

4) Intramolecular H – bond exist in O – nitrophenol but not in P – nitrophenol

5) SF_6 and XeF_6 are non - polar

49. How many of the following expressions and classifications are correct for the given equilibrium?

(i) $N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$ Homogeneous equilibrium

$K_C = \frac{[NO_2]^2}{[N_2O_4]}$ and $K_P = \frac{(P_{NO_2})^2}{(P_{N_2O_4})}$

(ii) $3Fe_{(s)} + 4H_2O_{(g)} \rightleftharpoons Fe_3O_{4(s)} + 4H_{2(g)}$

Hetereogeneous equilibrium $K_C = \frac{[H_2]^4}{[H_2O]^4}$ $K_P = \frac{(P_{H_2})^4}{(P_{H_2O})^4}$

(iii) $NH_4HS_{(s)} \rightleftharpoons NH_{3(g)} + H_2S_{(g)}$

Hetereogeneous equilibrium $K_C = [NH_3][H_2S]$ and $K_P = (P_{NH_3})(P_{H_2S})$

(iv) $CH_3COOH_{(l)} + C_2H_5OH_{(l)} \rightleftharpoons CH_3COOC_2H_{5(l)} + H_2O_{(l)}$

Homogeneous equilibrium

$K_C = \frac{[CH_3COOC_2H_5]\,[H_2O]}{[CH_3COOH]\,[C_2H_5OH]}$ but K_P is not define for liquid system

(v) $MgCO_{3(s)} \rightleftharpoons MgO_{(s)} + CO_{2(g)}$ Hetereogeneous equilibrium

$K_C = [CO_2]$ and $K_P = (P_{CO_2})$

(vi) $2H_2S_{(g)} \rightleftharpoons 2H_{2(g)} + S_{2(g)}$ Homogeneous equilibrium

$K_C = \frac{[H_2]^2\,[S_2]}{[H_2S]^2}$ and $K_P = \frac{(P_{H_2})^2\,(P_{S_2})}{(P_{H_2S})^2}$

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(vii) $SO_{2(g)} + NO_{2(g)} \rightleftharpoons SO_{3(g)} + NO_{(g)}$ Homogeneous equilibrium

$K_C = \frac{[SO_3]\ [NO]}{[SO_2][NO_2]}$ and $K_P = \frac{(P_{SO_3})\ (P_{NO})}{(P_{SO_2})(P_{NO_2})}$

(viii) $NH_4NO_{2(s)} \rightleftharpoons N_{2(g)} + 2H_2O_{(l)}$ Hetereogeneous equilibrium

$K_C = [N_2]$ and $K_P = (P_{N_2})$

50. How many of the following statements are correct

1) Adding an inert gas (like Argon) to a reaction mixture at equilibrium at **constant volume** will shift the equilibrium towards the side with more moles of gas.

2) Adding solid sodium acetate to an aqueous solution of acetic acid will increase the pH of the solution.

3) A buffer solution possesses its maximum buffering capacity when the pH of the solution is equal to the pKa of the weak acid component.

4) Pure water at 60^0C has a pH of approximately 6.5, which means it is acidic at this temperature. $(K_W = 10^{-13}M^2$ at $60^0C)$

5) The numerical value of the solubility product constant of a salt changes when a common ion is added to the saturated solution.

6) An aqueous solution of ammonium carbonate is basic

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MATHEMATICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. ABCD is a quadrilateral with $\overrightarrow{AB}=\vec{a}, \overrightarrow{AD}=\vec{b}, \overrightarrow{AC}=2\vec{a}+3\vec{b}$. If the area of the quadrilateral is λ times the area of parallelogram with $\overrightarrow{AB}$ and $\overrightarrow{AD}$ as adjacent side, then value of λ is:

(A) $\frac{3}{2}$ (B) $\frac{5}{2}$ (C) $\frac{7}{2}$ (D) 2

52. Shortest distance between the lines $\frac{x-3}{3}=\frac{y-8}{-1}=\frac{z-3}{1}$ and $\frac{x+3}{2}=\frac{y+7}{5}=\frac{z-6}{-1}$ is :

(A) $\sqrt{30}$ units (B) $3\sqrt{3}$ units (C) $4\sqrt{30}$ units (D) 0

53. Let P be the point of intersection of the lines $\frac{x-2}{1}=\frac{y-4}{5}=\frac{z-2}{1}$ and $\frac{x-3}{2}=\frac{y-2}{3}=\frac{z-3}{2}$. If the shortest distance of P from the line $4x=2y=z$ is k, then $7k^2$ is____

A) 54 B) 72 C) 18 D) 14

54. If $A=\begin{bmatrix}1 & 2 & -1\\ -1 & 1 & 2\\ 2 & -1 & 1\end{bmatrix}$, then $\det\left(adj\ A\right)=$

(A) 28 (B) 196 (C) 42 (D) 60

55. $\lim\limits_{x\to 0}\frac{2\sin x}{|x|}$ is equal to :

(A) –2 (B) 2 (C) 1 (D) limit doesn't exist

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56. If $A^k = O$ for some positive integer value of k and $(I-A)^p = I + A + A^2 + ... + A^{k-1}$, then p is

(A) – 1 (B) – 2 (C) – 3 (D) None of these

57. Equation of the line of shortest distance from P $(\bar{p})$ to the line $\bar{r} = \bar{a} + t\bar{b}$ is_____ (t, s are parameters)

(A) $\vec{r} = \vec{p} + s\left(\left(\vec{p}\times\vec{b}\right)\times\vec{b} - \left(\vec{a}\times\vec{b}\right)\times\vec{b}\right)$ (B) $\vec{r} = \vec{p} + s\left(\vec{p}\times\vec{b} + \vec{p}\times\vec{a}\right)$

(C) $\vec{r} = \vec{p} + s\left(\vec{p}\times\vec{b} + \vec{b}\times\vec{a}\right)$ (D) $\vec{r} = \vec{p} + s\left(\vec{a}\times\vec{b} + \vec{a} + \vec{b}\right)$

58. If $f(x) = \begin{cases} [\cos \pi x], & x < 1 \\ |x-2|, & 1 \le x < 2 \end{cases}$ ([*] denotes the greatest integer function), then $f(x)$ is

(A) continuous and non-differentiable at $x = -1$ and $x = 1$

(B) continuous at $x = \frac{1}{2}$

(C) discontinuous at $x = \frac{1}{2}$

(D) continuous but not differentiable at $x = 2$

59. The number of points where the function

$$f(x) = \begin{cases} \left(2x^2 - 3x - 7\right) & \text{if} \quad x \le -1 \\ \left[4x^2 - 1\right] & \text{if} \quad -1 < x < 1 \\ |x+1| + |x-2| & \text{if} \quad x \ge 1 \end{cases}$$

$[t]$ denotes the greatest integer $\le t$, is discontinuous is_____

(A) 5 (B) 14 (C) 8 (D) 7

60. If $f(x) = a\ln x + bx^2 + x$ has extremum at $x = 1$ and $x = 3$, then value of $(2b - a)$ is equal to

(A) 3 (B) $\frac{1}{2}$ (C) $\frac{3}{2}$ (D) $\frac{2}{3}$

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61. If $A=\cos(\cos x)+\sin(\cos x)$, then the greatest value of A is

(A) 1 (B) 2 (C) $\sqrt{2}$ (D) 0

62. The number of solutions of the equation $1+\sin x\sin^2\left(\frac{x}{2}\right)=0$ in $[-\pi,\pi]$ is

(A) 0 (B) 1 (C) 2 (D) 3

63. The coordinates of a point on the line $\frac{x-1}{2}=\frac{y+1}{-3}=z$ at a distance $4\sqrt{14}$ from the point $(1,-1,0)$ near to the origin are

(A) $(9,-13,4)$ (B) $\left(8\sqrt{4},-12,-1\right)$ (C) $\left(-8\sqrt{4},12,1\right)$ (D) $(-7,11,-4)$

64. If all the real values of m for which the function $f(x)=\frac{x^3}{3}-(m-3)\frac{x^2}{2}+mx-2013$ is strictly increasing in $x\in[0,\infty)$ is $[0,k]$, then the value of k is____

(A) 1 (B) 3 (C) 6 (D) 9

65. $\lim\limits_{x\to 2}\frac{\sqrt{(x+7)}-3\sqrt{(2x-3)}}{\sqrt[3]{(x+6)}-2\sqrt[3]{3x-5}}=$

(A) $\frac{31}{20}$ (B) $\frac{34}{23}$ C) $\frac{23}{31}$ D) $\frac{39}{17}$

66. The equation of a line through the point $(1,2)$ whose distance from the point $(3,1)$ has the greatest possible value is

(A) $x+y=3$ (B) $y=2x$

(C) $x+2y=5$ (D) $x-2y+3=0$

67. $\lim\limits_{n\to\infty}\frac{1^2+2^2+3^2+\ldots+n^2}{n^3}$ is equal to

(A) $\frac{1}{2}$ (B) $\frac{2}{3}$ (C) $\frac{1}{3}$ (D) $\frac{1}{6}$

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68. If f is differentiable function defined over R, $f(5)=7$ *and* $f'(5)=7$, then $\lim\limits_{x\to5}\dfrac{xf(5)-5f(x)}{x-5}$ is given by

(A) 35 (B) – 35 (C) 28 (D) – 28

69. The value of $\lim\limits_{x\to\infty}\dfrac{\sin x}{x}$, is

(A) 1 (B) 0 (C) – 1 (D) None of these

70. Let $f(x)=\dfrac{\sin 4\pi[x]}{1+[x]^2}$, where $[x]$ is the greatest integer less than or equal to x, then

(A) $f(x)$ is not differentiable at some points

(B) $f'(x)$ exists but is different from zero

(C) $f'(x)=0$ for all $x\in R$

(D) $f'(x)=0$ but f is not a constant function

SECTION – II

(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
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71. If *a, b, c* are real numbers and the matrix $A=\begin{bmatrix}a & b & c\\ b & c & a\\ c & a & b\end{bmatrix}$ is such that $A^TA=I$, then the value of $(a+b+c)^4+10$ is equal to : (where A^T is the transpose of A)

72. If $P(x)$ be a polynomial of degree 3, with $P(3)=0=P'(3)=P''(3)$. If $P'''(3)=12$, then value of $P(8)$ is equal to :

73. If A is a 2×2 matrix such that $trace(A)=det(A)=3$, then $trace\left(A^{-1}\right)$ is_____

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74. The least value of $\operatorname{cosec}^2 x + 25\sec^2 x, (x \in \text{domain of function})$ is_____

75. If $5a + 4b + 20c = t$, then value of 't' for which the line $ax + by + c - 1 = 0$ always passes through a fixed point is

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